

NAME: _____

PERIOD: _____

DATE: _____

Which Error Can the District Afford?

AP Statistics · Unit 3 · Topic 3.8 · B: 2026-12-15 · E: 2026-12-18

[STAR] TODAY'S PROBLEM

A district checks an emergency-alert system. Let p be the proportion of alerts that arrive late. It tests $H_0 : p = 0.05$ versus $H_a : p > 0.05$, replacing the system if it rejects H_0 . Both plans use $n = 200$. The Safety office wants to avoid keeping a system with a 10% late rate. The Budget office wants to avoid replacing a system with a 5% late rate. The offices have not agreed how to compare these costs.

Dev: “Use A. Smaller α lowers both error risks.”

Lin: “Use B. Its higher power has no cost when safety matters.”

Test plans ($n = 200$)

Plan	α	Power at $p = 0.10$
A	0.010	0.748
B	0.050	0.877

FIRST TAKE — before we discuss (5 min)

Which plan would you try first? Name the office you prioritized and one competing cost.

[BOOK] ERROR PROBABILITIES TRADE OFF (keep on the page)

A Type I error rejects a true H_0 , and its probability is α . A Type II error fails to reject a false H_0 ; $\beta = 1 - \text{power}$. With sample size fixed, increasing α generally raises power and lowers β , but it also raises the false-positive risk. Increasing sample size can raise power while keeping the chosen α fixed. Choose a plan by naming the real consequence of each error.

Work (a)–(c) from the rules box:

DOK 1 (a) Identify what the table's power at $p = 0.10$ means in this setting.

DOK 2 (b) Calculate the Type II error probability β for each plan. State each plan's Type I error probability.

DOK 3 (c) In 4–5 sentences, recommend a plan for one office using both error risks. Explain the consequence of each error and what Dev and Lin each overlook. State why the other office might choose differently.

Frame: For the _____ office, I recommend _____ because _____.

Frame: That plan risks _____ when _____; the other office may prioritize _____.

EXIT — turn this in

One thing the rules box changed about my first take: _____